

Yongchao Dang

Yongchao Dang (born 1992) earned his Doctor of Science (Technology) from Aalto University in 2023, a Master's degree from Xidian University in 2019, and completed an exchange year at Future University Hakodate, Japan. He obtained his Bachelor's degree from Northwestern Polytechnical University in 2016. He was a visiting researcher at EURECOM from January to April 2025. His research focuses on cellular-connected UAVs and the application of artificial intelligence and machine learning in wireless networks and O-RAN. He has hands-on experience implementing ETSI DECT-2020 NR with Nordic Semiconductor solutions, OpenWiFi with a self-developed radio intelligent controller for wireless TSN and Industry 5.0, and deploying O-RAN on OAI and srsRAN testbeds. He has contributed to several major EU Horizon 2020 flagship projects, including 5G!Drones and INSPIRE-5Gplus, as well as Finnish national projects such as DORIS-I/II and Cloudify-6G. He received the Nokia Scholarship (2022) and the IEEE Globecom Best Paper Award (2020) and has extensive experience collaborating with both industry and research institutes on practical R&D projects.

Education and Academic Background:

- **2023.10-now : Postdoc researcher, School of Electrical Engineering, Aalto University, Finland**
- **02/2025-05/2025: Visting scientist, EURECOM, Sophia Antipolis, France**
- **010/2020-10/2023: Doctor of Science (Technology), School of Electrical Engineering, Aalto University, Finland**
- **09/2016-06/2019: M.S. in Computer Science and Technology, XIDIAN University, Xi'an, China**
- **01/2018-10/2018: M.S. in School of Systems Information Science, Future University Hakodate, Hakodate, Japan**
- **09/2012-06/2016: B.S. in Computer Science and Technology, Northwestern Polytechnical University, Xi'an, China**

Research Interests:

- **5G/6G wireless networks for UAV and swarm communications & navigation**
- **GPS spoofing detection and mitigation for cellular-connected systems**
- **Agentic AI /ML techniques for network optimization, automation and security analytics**
- **Wireless Time-Sensitive Networking with Wi-Fi, DECT-2020 NR, 5G, and 6G**
- **Physical layer security and authentication, backscatter low-energy communication.**

Selected Publications:

Journals

- [1]. **Yongchao Dang**, Tao Chen, and Riku Jäntti, "AppWeaver: Agentic Discovery and Composition of O-RAN App Workflows for 6G," manuscript submitted to IEEE Wireless Communications, 2026.
- [2]. **Yongchao Dang**, Quang Ngo, Kalle Ruttik, Jose Costa Requena, and Riku Jäntti, "Uni-RAT: A Unified DECT-2020 NR and Wi-Fi System for 6G Multi-RAT Networks," manuscript submitted to Computer Networks, 2026.
- [3]. Zhang, Yifan, **Yongchao Dang**, Masoud Kaveh, Zheng Yan, Riku Jäntti, and Zhu Han. "Physical layer challenge-response authentication between ambient backscatter devices." IEEE Transactions on Information Forensics and Security (2026).
- [4]. **Dang, Yongchao**, Fagerholm Roni, Ruttik Kalle, Tao Chen, and Riku Jäntti. "Open Radio Intelligent Controller based Wireless Time Sensitive Networking for Industry 5.0." Authorea Preprints (2025).
- [5]. **Dang, Yongchao**, Ngo Quang, Fagerholm Roni, Rantanen Veli-Matti, Ruttik Kalle, and Riku Jäntti. "EWDC: Integrating DECT-2020 With Wi-Fi for Enhanced Wireless Direct Connectivity." IEEE Internet of Things Journal (2025).
- [6]. **Dang, Yongchao**, Alp Karakoc, Saba Norshahida, and Riku Jäntti. "3D radio map-based GPS spoofing detection and mitigation for cellular-connected UAVs." IEEE Transactions on Machine Learning in Communications and Networking 1 (2023): 313-327.
- [7]. Yang, Bin, **Yongchao Dang**, Tarik Taleb, Shikai Shen, and Xiaohong Jiang. "Sum rate and max-min rate for cellular-enabled UAV swarm networks." IEEE Transactions on Vehicular Technology 72, no. 1 (2022): 1073-1083.
- [8]. **Dang, Yongchao**, Chafika Benzaid, Bin Yang, Tarik Taleb, and Yulong Shen. "Deep-ensemble-learning-based GPS spoofing detection for cellular-connected UAVs." IEEE Internet of Things Journal 9, no. 24 (2022): 25068-25085.

Conference:

- [1]. Quang Ngo, **Yongchao Dang**, Kalle Koskinen, Kalle Ruttik, and Riku Jäntti, "Overhead-Optimized Short-Packet Communication for IIoT Using DECT NR+ Networks," 20th Biennial Baltic Electronics Conference (BEC2026), Tallinn, Estonia, 2026, accepted.
- [2]. Veli-Matti Rantanen, **Yongchao Dang**, and Kalle Ruttik, "TinyFlowCore: A Parameterized ASIP for Compact Protocol and Routing Applications," 20th Biennial Baltic Electronics Conference (BEC2026), Tallinn, Estonia, 2026, accepted.
- [3]. Ngo, Quang Quoc, Ruttik Kalle, **Dang Yongchao**, Jäntti Riku, and Jose Costa-Requena. "Toward Multi-Connectivity 6G IoT-Practical IoT Device-to-Core Integration." 6GNet, pp. 103-105. IEEE, 2025.

- [4]. Zhang, Yifan, **Yongchao Dang**, Yishan Yang, Masoud Kaveh, Zheng Yan, and Riku Jäntti. "AmbAu: Accurate and robust physical layer authentication for ambient backscatter devices." In ICC 2025-IEEE International Conference on Communications, pp. 1097-1102. IEEE, 2025.
- [5]. **Dang, Yongchao**, Alp Karakoc, and Riku Jäntti. "Graphic Neural Network based GPS Spoofing Detection for Cellular-Connected UAV swarm." In 2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring), pp. 1-6. IEEE, 2023.
- [6]. **Dang, Yongchao**, Chafika Benzaïd, Tarik Taleb, Bin Yang, and Yulong Shen. "Transfer learning based GPS spoofing detection for cellular-connected UAVs." In 2022 International Wireless Communications and Mobile Computing (IWCMC), pp. 629-634. IEEE, 2022.
- [7]. **Dang, Yongchao**, Chafika Benzaïd, Bin Yang, and Tarik Taleb. "Deep learning for GPS spoofing detection in cellular-enabled UAV systems." In 2021 International Conference on Networking and Network Applications (NaNA), pp. 501-506. IEEE, 2021.
- [8]. **Dang, Yongchao**, Chafika Benzaïd, Yulong Shen, and Tarik Taleb. "GPS spoofing detector with adaptive trustable residence area for cellular based-UAVs." In GLOBECOM 2020-2020 IEEE Global Communications Conference, pp. 1-6. IEEE, 2020.
- [9]. Wu, Huihui, Xuening Liao, **Yongchao Dang**, Yulong Shen, and Xiaohong Jiang. "Limits of covert communication on two-hop AWGN channels." In 2017 international conference on networking and network applications (NaNA), pp. 42-47. IEEE, 2017.
- [10]. **Dang, Yongchao**, Yin Chen, Huihui Wu, Yulong Shen, and Xiaohong Jiang. "Physical layer authentication and identification in wireless network via the locations of surrounding noise sources." In 2017 International Conference on Networking and Network Applications (NaNA), pp. 30-35. IEEE, 2017.

Research and Project Experience:

- **AURORA-6G: AI-driven Unified O-RAN for Real-time Autonomous Zero-touch Networks (Just release)**
- **6GARROW – 6G AI-Native Integrated RAN-Core Networks**
(EU –SNS: Finland and Republic of Korea joint project, 01/01/2025 → 31/12/2027)
 - Investigating CU-UP integration with the core UPF to support edge-based URLLC.
 - Responsible for architecture review and design of the integrated RAN-core solution.
- **Cloudify-6G – Data-Driven Distributed Cloud & Communication System**
(Business Finland, 05/2023 – 04/2026)
 - Deliverable: D1.1 use-case analysis,
 - Deliverable: D2.1 radio-protocol components & design.
 - Deliverable: D4.1 Cloudify platform development focused on: IS-Wireless O-RAN, OAI Flex-RIC, srsRAN-RIC.
 - Demonstration: Developed and tested ORAN KPM and near-real time xApps for RAN slicing and resources scheduling.
- **6G-eMTC – Extreme Machine-Type Communications for 6G**
(Business Finland, 01/2023 – 12/2024)
 - Demonstration: Developing DECT-2020 NR MAC layer with Nordic NRF9161 chip set.
 - Publication: EWDC: Integrating DECT-2020 with Wi-Fi for Enhanced Wireless Direct Connectivity
- **5G-TSN – Wireless TSN over Programmable UPF**
(Visiting EURECOM, France, 02/2025 – 04/2025)
 - Demonstration: Mapping TSN QoS to 5G QFI with different RL and other methods.
 - EURECOM work review on 5G radio protocols stack design and implementation for RAN slicing and TSN support.
- **Drolo-I / Drolo-II – GPS Spoofing Detection & Mitigation**
(EU Recovery and Resilience Facility, 01/09/2023 → 31/08/2025)
 - Research and development of advanced spoofing-resistant positioning techniques.
 - Network measurements of cellular-connected UAVs.
- **INSPIRE-5Gplus: INtelligent Security and Pervasive tRust for 5G and Beyond**
(EU Horizon 2020, 01/11/2019 → 31/10/2022)
 - Designed intelligent security and trust mechanisms for 5G networks.
 - Enabler: UAV position monitoring and anti GPS Spoofing
 - Deliverable: D2.1: 5G Security: Current Status and Future Trends
 - Deliverable: D3.1: 5G security assets baseline and advancements

- **TERAWAY:** Terahertz technology for ultra-broadband and ultra-wideband operation inside 5G networks with SDN management of network and radio resources
(EU Horizon 2020, 01/11/2019 → 31/08/2023)
- Radio map construction, deterministic channel modelling, ray tracing tool design.
- Publication: 3D radio map-based GPS spoofing detection and mitigation for cellular-connected UAVs
- **5G!Drones:** Unmanned Aerial Vehicle Vertical Applications' Trials Leveraging Advanced 5G Facilities
(EU Horizon 2020, 01/06/2019 → 30/11/2022)
- Designed and developed a 5G-connected UAV system leveraging Aalto's 5G network infrastructure.
- Publications in the list related to cellular-connected UAVs
- PHD thesis: ML based GPS spoofing detection and mitigation for cellular connected UAVs

Awards and Scholarships:

- Aalto University Doctoral Incentive Scholarship (2023): (€2,000)
- Nokia Scholarship (2022): Awarded for research on machine learning-based GPS spoofing detection (€5,000)
- Aalto University Incentive Scholarship (2022): (€1,000)
- Best Paper Award, IEEE Globecom (2020): For “GPS Spoofing Detector with Adaptive Trustable Residence Area for Cellular-based UAVs”

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